# MSG No. 114.278 $P\bar{4}'2_1c'$ [ Type III, tetragonal ]

Table 1: Wyckoff site: `2a`, site symmetry: `-4'..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ 0]$ | `[1,4,5,6]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[2,3,7,8]` |

Table 2: Wyckoff site: `2b`, site symmetry: `-4'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ 0,\ \frac{1}{2}\right]$ | `[1,4,5,6]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$ | `[2,3,7,8]` |

Table 3: Wyckoff site: `4c`, site symmetry: `2..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | `[1,4]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | `[2,3]` |
| 3 | $[0,\ 0,\ -z]$ | `[5,6]` |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[7,8]` |

Table 4: Wyckoff site: `4d`, site symmetry: `2..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ z\right]$ | `[1,4]` |
| 2 | $\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$ | `[2,3]` |
| 3 | $\left[\frac{1}{2},\ 0,\ -z\right]$ | `[5,6]` |
| 4 | $\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[7,8]` |

Table 5: Wyckoff site: `8e`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | `[1]` |
| 2 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$ | `[2]` |
| 3 | $\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$ | `[3]` |
| 4 | $[-x,\ -y,\ z]$ | `[4]` |
| 5 | $[y,\ -x,\ -z]$ | `[5]` |
| 6 | $[-y,\ x,\ -z]$ | `[6]` |
| 7 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$ | `[7]` |
| 8 | $\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$ | `[8]` |